

Hidden Markov Model: Statistical Analysis Steps

27/07/2026

Study Design

An HMM based on a retrospective longitudinal cohort of children born 2012-2019 with continuous residence in Lothian since birth.

Aim and objectives

To quantify the contribution of underheated homes to preschool ARI's by:

1. Developing the infrastructure for multilevel count models and latent state models.
2. Examining associations between home energy efficiency and ARI burden in 0 to 5-year-olds.

Exposure

At risk of living in a cold house – using the initial definition of: EPC rating $\leq D$ and postcode level (median household) income is $< 60\%$ of the Scottish median household income after housing costs (£34,258 based on 2022-2025 figures)¹. This equates to: (EPC $\leq D$) + (PC income $\leq$ £20,555), adjusted annually for inflation² - *Appendix 1*.

Outcome

Respiratory health state.

Covariates

Individual level: Sex, Ethnicity, Maternal smoking status, Comorbidities

Household level: Tenure, Floor area,

Small area level: Climate & Pollution (time-dependent), SIMD

Components

States (S)

1. Start with 3 states: Good Respiratory Health, Poor Respiratory Health, Very Poor Respiratory Health.
2. An individual's State (s_i) will be inferred from the observed variable/s (z_i) for each time point (t_i).

Observations (Z)

1. Prescriptions dispensed for respiratory illness (≥ 1 per time point).
2. Hospital admissions for respiratory illness (≥ 1 per time point) .
3. Mortality resulting from respiratory illness.

Time points (t)

1. Start point (t_0) aligns with birth.
2. Time points accrue ($t_0 \rightarrow t_1 \rightarrow \dots t_n$) at one-month intervals.
3. End point (t_n) aligns with age five-years, end of the study period or death
4. Censoring time (t_c) aligns with the age at which an external event terminates the observation sequence (migration from Lothian or mortality). At t_c the individual is excluded from the dataset, based on the assumption that the censoring mechanism is independent of future states (*uninformed censoring*).

¹ <https://ifs.org.uk/publications/labour-market-living-standards-and-poverty-trends-scotland>

² <https://www.bankofengland.co.uk/monetary-policy/inflation/inflation-calculator>

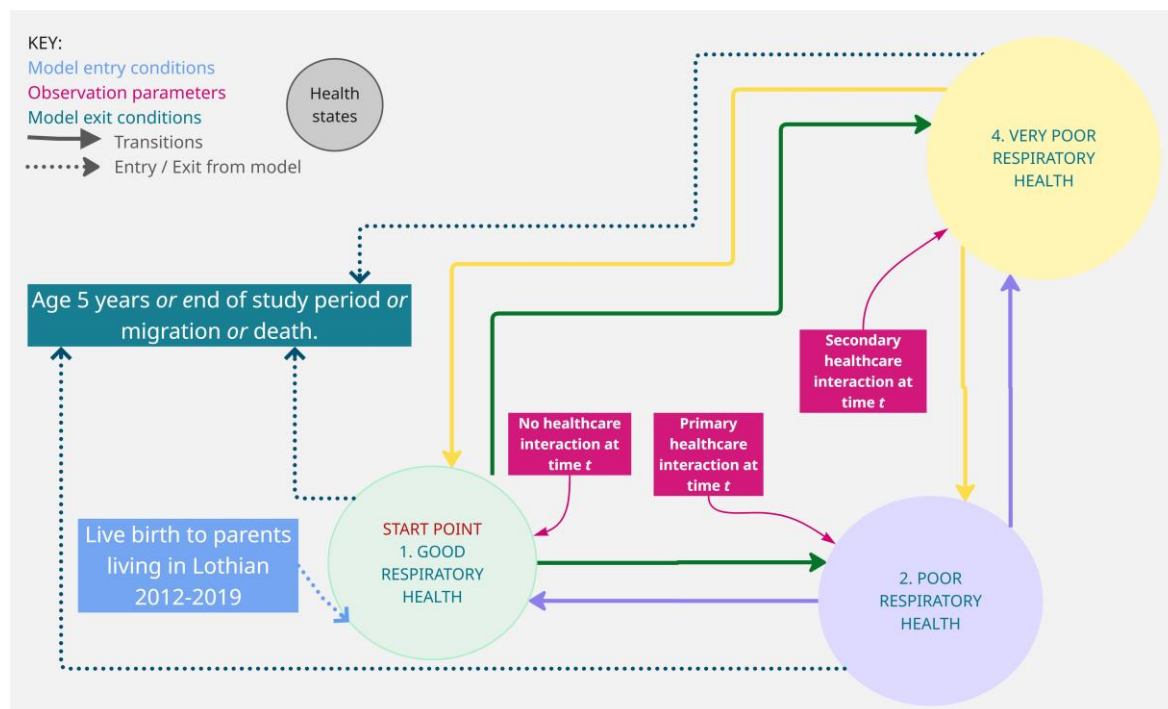
Transition probability Matrix (Q)

1. Transition probabilities will be based on the proportion of individuals in state (a) at t_j that transition to state (b) at t_{j+1} : i.e.: $P_{ab} = N_{ab}/N_a$.
2. Transitions probabilities will be derived from the emission distribution of (Z) at t_j , assuming that these emissions are independent from observations during t_{j-1} .
3. All possible transition probabilities, including remaining in the same state will be included in matrix Q (3x3 for a 3-state model).
4. The sum of all possible transitions from a state, a row in Q, i.e. (a->a, a->b, a->c), =1.

Random effects (R)

Random effects will be derived from covariates (listed above). Final selection will be based on a DAG to determine their relationship with the exposure and outcome, having taken collinearity into consideration. Priors will assume a zero mean normal distribution $R_j \sim \text{Normal}(0, \sigma_r^2)$, but with a wider dispersion.

Summary of components and their relationships



Data requirements

Repeat measures are required with one row for each time point. Individuals are differentiated using a UID. Likelihood of the full dataset is computed as the product of the likelihoods of all individuals (assuming independence).

Random effects should be formatted as factors – not character strings or integers

Missing values in response variables should be noted as NA.

Missing values in covariates need to be interpolated. The default, for NA values is to assign: (1) the last non-missing value, (2) the next non-missing value- where there are no previous values provided.

Analysis

Analysis will be conducted in R using the Bayesian inference package `tmb Stan`. Markov Chain Monte Carlo (MCMC) will be used to sample over random effects with (`HMM$fit_stan()`), to estimate their *conditional posterior distribution*, fitted using Hamiltonian Monte Carlo.

The model will be constructed from three classes:

1. **Markov Chain class** (`MarkovChain$new()`): consisting of the dataset (`data`), initial state, assuming healthy at birth (`initial.state`), the number of states - assume 3 (`n.states`), transition probability matrices (`tpm`) and associated formulae (`formulas`) tpm formulae will be informed by a combination of raw Dataloch and PICNIC values.
2. **Observation class** (`Observation$new()`): consisting of observed variables (`data`), their distribution (`dists`)- for example normal, poisson etc. depending on how the observation has been classified, the number of states - assume 3 (`n.states`), formulae (`formulas`) where a variable has dependence on other covariates and initial parameter values (`par`) for variables.
3. **HMM class** (`HMM$new()`) combining the Markov Chain and Observational classes to perform statistical analysis.

Priors will be specified for fixed effects (α) and smoothness parameters (λ) (for splines) using (`HMM$set.priors`). The model will then be fitted (`HMM$fit_stan()`) – stating the number of chains and the number of iterations within each chain stated. These will be determined by numbers required to achieve convergence. This will be tested using a subset of the data, with a minimum starting point of 3 chains and 1,000 iterations.

Models will be constructed using progressive levels of exposure to living in a cold house with transition probabilities and sojourn time calculated for each model. Hazard ratios will be used to quantify the degree to which risk of *living in a cold house*, impacts on ARI in children, with children at *least risk of living in a cold house* used as the reference population. Hazard ratios (Cox proportional hazard) and associated confidence intervals will be calculated using the `survival` package, `coxph` and `mstate` extensions – to handle transition between multiple states.

Outputs

Outputs will be obtained using `HMM$out_stan()` and inspected using `as.matrix` to inspect posterior distribution as well as using density plots `rstan::stan_dens()`.

`HMM$viterbi()` and `HMM$plot_ts()` will be used to visualise the distribution of the states over. `HMM$plot()` will be used to investigate transition probabilities between states. Variance will be reported in standard deviations using confidence intervals for plots.

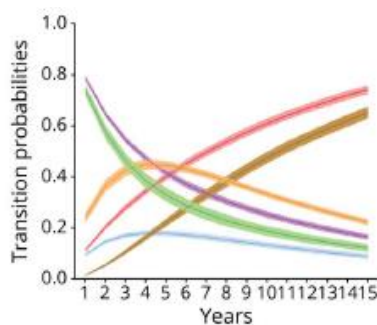
Hazard ratio's will be outputted as a table.

Visualisation

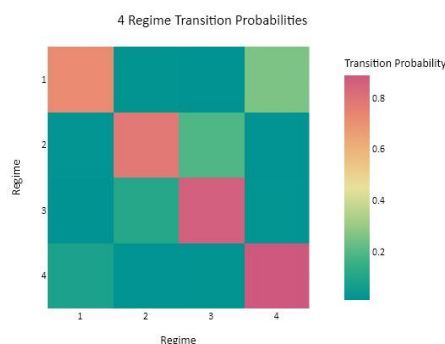
Visualisation will be conducted using [bayesplot](#) and [ggplot2](#).

Splines: the transition probabilities listed will be mapped against the time variable (age) as B-Splines using [splines](#) and [ggplot2](#):

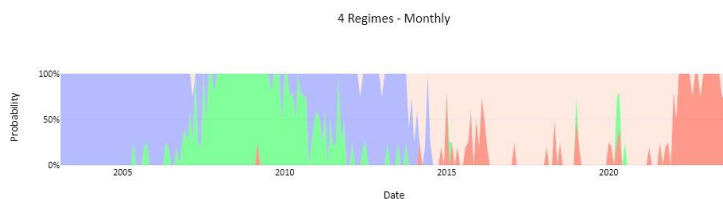
- Healthy -> Sick
- Healthy -> Very Sick
- Sick -> Healthy
- Sick -> Very Sick
- Very Sick -> Sick
- Very Sick -> Healthy



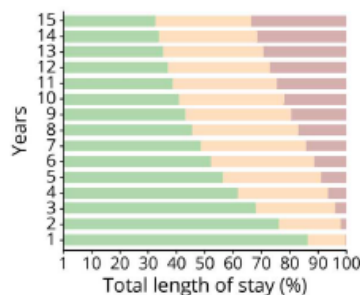
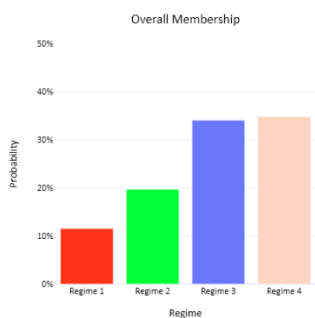
Heatmap: the transition probability for the whole follow-up period visualised as a heatmap using [geom_tiles](#), with a colour gradient used to illustrate most likely transitions.



Proportional graph: proportion of children in each state at each timepoint will be categorized using the Viterbi output and plotted using [geom_area](#).



Bar graphs: Mean sojourn time in each state across the whole follow-up period will be plotted using [geom_histogram](#) or [geom_bar](#).



Appendix 1: Relative poverty line AHC Scotland.

To calculate exposure risk: *living in a cold house*, base year 2025, rate calculated from 2022-2025 IFS statistic and adjusted for inflation.

Year	Adjustment	Annual adjustment	Value
2012	-30.6%	-2.69%	£14,263
2013	-28.8%	-2.70%	£14,629
2014	-27.8%	-2.81%	£14,843
2015	-27.8%	-3.97%	£14,849
2016	-27.3%	-3.33%	£14,947
2017	-25.3%	-3.41%	£15,348
2018	-23.5%	-3.54%	£15,728
2019	-22.1%	-3.82%	£16,010
2020	-21.4%	-4.36%	£16,146
2021	-19.4%	-4.77%	£16,564
2022	-12.1%	-3.70%	£18,071
2023	-4.5%	-1.89%	£19,630
2024	-3.3%	-2.34%	£19,876
2025	/	/	£20,555